

Kristen Marie Johnson

RESEARCH INTERESTS

My research develops machine learning and natural language processing (NLP) models for the understanding, prediction, and mitigation of bias and stereotypes, and for moral self-correction (alignment) and value acquisition in NLP models, specifically LLMs. Our research lab focuses on ethics in social computing via bias and morality prediction and mitigation from two perspectives: through application of social science theories in combination with mathematically-guided techniques, e.g., parameter estimation and fine-tuning. My past research in NLP focused on the analysis of political discourse on social media and in the media. My work consisted of the prediction and classification of political stances, policy framing techniques, and underlying moral frameworks, and their application for political behavior prediction.

EDUCATION

Ph.D. in Computer Science August 2019

Purdue University, West Lafayette, IN
Dissertation: Modeling Language, Social, and Behavioral Abstractions for Microblog Political Discourse Classification
Advisor: Dr. Dan Goldwasser

Master of Science in Computer Science May 2012

University of New Orleans, New Orleans, LA GPA: 3.9
Thesis: Software for Estimation of Human Transcriptome Isoform Expression Using RNA-Seq Data
Advisor: Dr. Dongxiao Zhu

Bachelor of Science in Psychology May 2007

University of New Orleans, New Orleans, LA GPA: 3.9
Magna Cum Laude

PUBLICATIONS

Diagnosing Moral Reasoning Acquisition in Language Models: Pragmatics and Generalization. G. Liu, Z. Qi, X. Zhang, L. Jiang, **K. Johnson**.

In *Findings of the Association for Computational Linguistics: EMNLP*, 2025.

Discourse Heuristics For Paradoxically Moral Self-Correction.

G. Liu, Z. Qi, X. Zhang, **K. Johnson**.

In *Findings of the Association for Computational Linguistics: EMNLP*, 2025.

A Survey to Recent Progress Towards Understanding In-Context Learning.

H. Mao, G. Liu, Y. Ma, R. Wang, **K. Johnson**, J. Tang.

In *Findings of the Association for Computational Linguistics: NAACL*, 2025. [PDF]

Moral Self-correction Is Not An Innate Capability of Large Language Models.

G. Liu, Z. Qi, L. Cheng, and **K. Johnson**.

To appear in *EMNLP Findings*, 2025. [PDF]

On the Intrinsic Self-Correction Capability of LLMs: Uncertainty and Latent Concept.

G. Liu, H. Mao, B. Cao, Z. Xue, X. Zhang, R. Wang, J. Tang, and **K. Johnson**.

To appear in *EMNLP Findings*, 2025. [PDF]

Korean Stereotype Content Model: Translating Stereotypes Across Cultures.

M. Kim and **K. Johnson**.

In *Proc. of the 3rd Workshop on Cross-Cultural Considerations in NLP (C3NLP 2025)*. [PDF]

No Free Lunch for Defending Against Prefilling Attack by In-Context Learning.
G. Liu, Z. Xue, B. Chen, **K. Johnson**, and R. Pedarsani.
arXiv preprint arXiv:2412.12192, 2024. [PDF]

Smaller Large Language Models Can Do Moral Self-Correction.
G. Liu, Z. Xue, R. Wang, and **K. Johnson**.
In *Proc. of the 5th Workshop on Trustworthy NLP (TrustNLP)*, 2025. [PDF]

Intrinsic Self-correction for Enhanced Morality: An Analysis of Internal Mechanisms and the Superficial Hypothesis. G. Liu, H. Mao, J. Tang, and **K. Johnson**.
In *Proc. Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024. [PDF]

Towards Understanding Task-agnostic Debiasing Through the Lenses of Intrinsic Bias and Forgetfulness.
G. Liu, M. Afshari, X. Zhang, Z. Xue, A. Ghosh, B. Bashyal, R. Wang, and **K. Johnson**.
In *Findings of the Association for Computational Linguistics (ACL)*, 2024. [PDF]

PAC-tuning: Fine-tuning Pretrained Language Models with PAC-driven Perturbed Gradient Descent.
G. Liu, Z. Xue, X. Zhang, **K. Johnson**, R. Wang.
In *Proc. of the 2024 Conference on Empirical Methods for Natural Language Processing, EMNLP*, 2025.

ABLE: Agency-BeLiefs Embedding to Address Stereotypical Bias through Awareness instead of Obliviousness. M. Kim, J. Kim, and **K. Johnson**.
In *Proc. Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*, 2024. [Link]

ABLE: Agency-BeLiefs Embedding to Address Stereotypical Bias through Awareness instead of Obliviousness. M. Kim, J. Kim, and **K. Johnson**. Midwest Speech and Language Days (MSLD), 2024.

Race, Gender, and Age Biases in Biomedical Masked Language Models.
M. Kim, J. Kim, and **K. Johnson**.
In *Findings of the Association for Computational Linguistics (ACL)*, 2023. [Link]

ASDClaims: Twitter Dataset of Claims on Autism Spectrum Disorder.
M. Kim, J. Kim, B. Kim, **K. Johnson**, and J. Kim.
In *Proc. BDA4HM 2022: IEEE International Conference on Big Data Workshop*, 2022.

CLoSE: Contrastive Learning of Subframe Embeddings for Political Bias Classification of News Media. M. Kim and **K. Johnson**.
In *Proc. COLING, 29th International Conference on Computational Linguistics*, 2022. [Link] [PDF]

Cryptocurrency Day Trading and Framing Prediction in Microblog Discourse.
A. Pawlicka Maule and **K. Johnson**.
In *Proc. Third Workshop on Economics and Natural Language Processing (EcoNLP, in conjunction with EMNLP'21)*, 2021.

Using Social Media For Bitcoin Day Trading Behavior Prediction.
A. Pawlicka Maule and **K. Johnson**.
In *Proc. Fourth Workshop for Widening NLP (WiNLP, in conjunction with ACL'20)*, 2020. [Link]

Modeling Language, Social, and Behavioral Abstractions for Microblog Political Discourse Classification. **K. Johnson**.

Purdue University Graduate School Dissertations, 2019. [\[Link\]](#) [\[PDF\]](#)

Modeling Behavioral Aspects of Social Media Discourse for Moral Classification.

K. Johnson and D. Goldwasser.

In Proc. *Third Workshop on NLP and Computational Social Science (NLP+CSS, in conjunction with NAACL'19)*, 2019. [\[Link\]](#) [\[PDF\]](#)

Classification of Moral Foundations in Microblog Political Discourse.

K. Johnson and D. Goldwasser.

In Proc. *56th Annual Meeting of the Association for Computational Linguistics (ACL'18)*, 2018. [\[Link\]](#) [\[PDF\]](#)

Modeling of Policy Frames for Morality Detection on Twitter.

K. Johnson and D. Goldwasser.

In Proc. *Second Workshop on Women and Underrepresented Minorities in NLP (WiNLP, in conjunction with NAACL'18)*, 2018. [\[Link\]](#)

Leveraging Behavioral and Social Information for Weakly Supervised Collective Classification of Political Discourse on Twitter.

K. Johnson, D. Jin, and D. Goldwasser.

In Proc. *55th Annual Meeting of the Association for Computational Linguistics (ACL'17)*, 2017. [\[Link\]](#) [\[Talk\]](#) [\[PDF\]](#)

Ideological Phrase Indicators for Classification of Political Discourse Framing on Twitter.

K. Johnson, I. Lee, and D. Goldwasser.

In Proc. *Second Workshop on NLP and Computational Social Science (NLP+CSS, in conjunction with ACL'17)*, 2017. [\[Link\]](#) [\[PDF\]](#)

Modeling of Political Discourse on Twitter.

K. Johnson and D. Goldwasser.

In Proc. *First Workshop on Women and Underrepresented Minorities in NLP (WiNLP, in conjunction with ACL'17)*, 2017.

PurdueNLP at SemEval-2017 Task 1: Predicting Semantic Textual Similarity with Paraphrase and Event Embeddings.

I. Lee, M. Goindani, C. Li, D. Jin, **K. Johnson** X. Zhang, M. Pacheco, D. Goldwasser.

In Proc. *11th International Workshop on Semantic Evaluation (SemEval-2017, in conjunction with ACL'17)*, 2017. [\[Link\]](#) [\[PDF\]](#)

Modeling of Political Discourse Framing on Twitter.

K. Johnson, D. Jin, and D. Goldwasser.

In Proc. *11th International AAAI Conference on Web and Social Media (ICWSM-17)*, 2017. [\[Link\]](#) [\[PDF\]](#)

“All I know about politics is what I read in Twitter”: Weakly Supervised Models for Extracting Politicians’ Stances from Twitter.

K. Johnson and D. Goldwasser.

In Proc. *COLING 2016, 26th International Conference on Computational Linguistics*, 2016. [\[Link\]](#) [\[PDF\]](#)

Identifying Stance by Analyzing Political Discourse on Twitter.

K. Johnson and D. Goldwasser.

In Proc. *First Workshop on NLP and Computational Social Science (NLP+CSS, in conjunction with EMNLP'16)*, 2016. [\[Link\]](#) [\[PDF\]](#)

Software for Estimation of Human Transcriptome Isoform Expression Using RNA-Seq Data.

K. Johnson.

University of New Orleans Theses and Dissertations, 2012. [\[Link\]](#) [\[PDF\]](#)

SPATA: A Seeding and Patching Algorithm for De-Novo Transcriptome Assembly.

Z. Zhao, T. Nguyen, N. Deng, **K. Johnson**, and D. Zhu.

IEEE International Conference on Bioinformatics and Biomedicine Workshops (BIBM), 2011.

[\[Link\]](#)

Isoform-Level microRNA-155 Target Prediction Using RNA-Seq.

N. Deng, A. Puetter, K. Zhang, **K. Johnson**, Z. Zhao, C. Taylor, E. Flemington, and D. Zhu.

Nucleic Acids Research, 39(9), 2011. [\[Link\]](#) [\[PDF\]](#)

INVITED TALKS AND PANELS

Panel: Women in Technology.

Novi High School Chapter of Girls Who Code, March 2022.

Modeling and Understanding Framing in Microblog Political Discourse.

Colloquium on Narrative Framing and Its Linguistic Forms in Online Media,

Communication in Multicultural Society Conference (CMSC), Dec. 2020.

Panels: Graduate School Advice and Careers in Academia and Industry.

ACL Mentoring Panels, July 2020.

Modeling of Language, Social, and Behavioral Abstractions for Microblog Political Discourse Classification.

Graduate Seminar, Wayne State University, Sept. 2020.

Practical Data Science with Applications in Industry, INFORMS Annual Meeting, Oct. 2019.

Natural Language Processing Methods for Extracting Political Behavior from Twitter.

Midwest Speech and Language Days and Midwest Computational Linguistics Colloquium, 2016.

RESEARCH

Assistant Professor
GenCen Consulting and
GJEC Associated Faculty
Dept. of Computer Science and
Engineering

August 2019 to Present
Michigan State University

My current research aims to expand my abstraction modeling techniques for other social science applications, including studying politics over time, social justice, legislation, and their associated real-world repercussions. I am also exploring new ways to model microblog discourse for financial and sociological domains. In addition to research, I teach the graduate level Natural Language Processing course and undergraduate Introduction to Machine Learning course.

Research Assistant

May 2015 to July 2019

Dr. Dan Goldwasser, Purdue NLP Group

Purdue University

My research with Professor Goldwasser explored the analysis of political discourse and its impacts on real-world behavior. Using Probabilistic Soft Logic (PSL), I modeled the language, social behaviors, and framing strategies of politicians and political groups on Twitter to predict real-world political actions and analyze trends around political and social events.

Research Assistant

May 2010 to May 2012

Dr. Dongxiao Zhu and Dr. Zhiyu Zhao

University of New Orleans

My research with Professor Zhu focused on development of the software packages: GWIE, iGWIE, and ChromIE. This software was a global extension of our group's novel EM-like algorithm to allow genome-wide isoform level estimation of the human transcriptome from RNA-Seq data. Our work was performed in collaboration with Children's Hospital and Tulane University of New Orleans, LA. I also worked in collaboration with Dr. Zhao to develop and test software for de-novo transcriptome assembly and protein docking.

TEACHING	Assistant Professor CSE 404: Introduction to Machine Learning CSE 842: Natural Language Processing	Michigan State University
	Graduate Teaching Assistant CS 590-NLP: Machine Learning Methods for NLP CS 240: Programming in C CS 565: Programming Languages CS 180: Introduction to Object Oriented Programming	Purdue University January 2016 to May 2016 June 2014 to August 2014 January 2014 to May 2014 August 2013 to December 2013
MENTORING	MSU Professorial Assistant Advisor	Fall 2022 - Fall 2024
	EMNLP Mentoring Program	November 2021
	EMNLP Computational Social Science Birds of a Feather	November 2021
	ACL Diversity & Inclusion Mentor	July 2020
	NAACL Big Sibling Mentor	June 2019
	Purdue Women in Science Programs (WISP)	August 2015 to March 2017
SERVICE	NSF Panel Member	
	Human Language Technologies, Small and CRII Proposal Reviewer	2023
	Area Chair	
	Conference on Empirical Methods in Natural Language Processing (EMNLP)	2021, 2023
	Social Media and Sponsorships Co-chair	
	Widening Natural Language Processing Workshop (WiNLP)	2019
	Local Sponsorships and Exhibits Chair	
	North American Chapter of the Association for Computational Linguistics:	
	Human Language Technologies (NAACL)	2018
	Conference Reviewer & Meta-Reviewer	
	ACL Rolling Review Member	2021 - 2025
	LREC-COLING	2024
	North American Chapter of the Association for Computational Linguistics (NAACL)	2021
	European Chapter of the ACL (EACL)	2021 - 2023
	The SIGNLL Conference on Computational Natural Language Learning (CoNLL)	2021 - 2022
	AAAI Conference on Artificial Intelligence (AAAI)	2019 - 2023
	Annual Meeting of the Association for Computational Linguistics (ACL)	2018 - 2023
	Conference on Empirical Methods in Natural Language Processing (EMNLP)	2018 - 2023
	Workshop Reviewer	
	Workshop on Noisy User-generated Text (W-NUT)	2018 - 2025
	Workshop on NLP and Computational Social Science (NLP+CSS)	2017 - 2022
	Journal Reviewer	
	Transactions of the Association for Computational Linguistics (TACL)	2025
	Journal of Artificial Intelligence Research (JAIR)	2020
	MSU Department of Computer Science and Engineering	
	Graduate Studies and Research Committee Member	2019 - 2023

AWARDS

Travel Grants

Widening Natural Language Processing Workshop (WiNLP)	2018
International AAAI Conference on Web and Social Media (ICWSM)	2017
Workshop on NLP and Computational Social Science (NLP+CSS)	2016, 2019
Purdue University Women in Science Programs (WISP)	2016

MEDIA COVERAGE

Cryptocurrency Day Trading Prediction.

Wealth Simple Magazine. Spring 2022.

Artificial Intelligence.

Insights Magazine. Tim Brouk. Spring 2017.

Politicians on Twitter: What Can You Make of It?

Purdue Policy Research Institute (PPRI).